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ORIGINAL PAPER

Workspace analysis of an underconstrained cable-driven parallel robot with four cables

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Abstract

A cable-driven parallel robot is controlled by varying the cable lengths. When the moving platform of an underconstrained cable robot, suspended using four cables and acted upon by only the cable forces and weight of the platform, is moved to a particular position in a workspace, it can have only a limited variation in orientation. The feasible range of orientations at different positions in a cuboidal workspace is obtained in this work. It is not attempted to find the poses where the mobile platform is in complete static equilibrium. The range of poses, when the net force in all directions and the moment of forces about vertical axis are all equal to zero, and the moments about two horizontal axes are limited to specified limits, are mapped. Static simulation of the cable-driven robot at different positions along a prescribed path, as it transfers a payload, is conducted. The set of cable lengths corresponding to a position and orientation is given as input to the simulation and the equilibrium position is found. It is observed that when the cable lengths correspond to a feasible orientation, the moving platform stays on the intended path and when cable lengths corresponding to infeasible orientation are given as input, the center of the platform drifts away from the desired position. Knowledge of the feasible range of orientations will be useful in avoiding this drift and guiding the platform along a desired path.

Keywords Cable-driven parallel robots · Underconstrained robots · Parallel mechanisms · Workspace · Stability analysis

1 Introduction

A cable-driven parallel robot (CDPR) has a moving platform which is suspended using cables. The end effector is attached to or mounted on this platform. The pose of the end effector is changed by varying the lengths of the cables. One of the pioneering models of cable suspended robots was introduced by Albus et al. (Albus et al. 1992). It was conceived as a robot crane which could lift, maneuver and position large loads. Compared to robots with rigid links, cable-driven robots can have very large workspaces. As the weight of cable is negligible, the payload to weight ratio is also very high and hence they are suitable for high acceleration applications (Bosscher and Ebert-Uphoff 2004; Diao and Ma 2007). Cable-driven robots have been suggested for wide range of uses that include pick and place operations, rescue operations (Albus et al. 1992; Bosscher and Ebert-Uphoff 2004; Mottola et al. 2019), and assisting in human rehabilitation exercises (Hamida et al. 2021). Use of CDPR in guiding concrete extruder in counter crafting (Bosscher et al. 2007) and in large scale 3D printing of structures (Barnett and Gosselin 2015; Izard et al. 2017) have been demonstrated.

A CDPR can be either a fully constrained one or an underconstrained one. The number of cables required to fully constrain the cable driven mechanism is $n+1$, where n is the degrees of freedom (Roberts et al. 1998). Thus four cables are required to fully constrain a planar mechanism and seven cables for a spatial mechanism. In fully constrained CDPRs the position and orientation of the end effector is fully defined when lengths of cables are specified (Bosscher and Ebert-Uphoff 2004). The fully constrained mechanism has one obvious disadvantage–the cables interfere with objects in the workspace. In underconstrained version of the CDPR, this disadvantage is reduced considerably, but at the cost of another—that of reduced stability (Bosscher and Ebert-Uphoff 2004). For any given set of lengths for the cables, the pose of the platform is determined by the gravity force, external wrench and the inertia forces (Carricato and Merlet 2013). A discussion on the stability of

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underconstrained cable robots can be seen in Bosscher and Ebert-Uphoff (Bosscher and Ebert-Uphoff 2004), where, a measure of stability based on the slope of the constrained surface formed by the cables, is developed.

While applying additive manufacturing methods for buildings and structures, cable-driven robots have several advantages over gantry-type Cartesian robots and articulated robotic arms, as positioning devices for the material depositing head. One is, installing a CDPR to cover the large workspace corresponding to a building is much cheaper compared to building a rigid-link robot. Others being, the easiness with which a CDPR can be re-configured for a different sized structure and the easiness in transporting the mechanism to another site (Bosscher et al. 2007; Izard et al. 2017). Bosscher et al. propose the use of CDPR for building the structure of an entire building using contour crafting, with the end effector consisting of concrete extrusion nozzle and trowelling tool for shaping. The authors selected fully constrained CDPR for their study due to the high stiffness it ensures, but, report to have considered the choice of underconstrained CDPR also (Bosscher et al. 2007). Underconstrained CDPR has been successfully used to position the foam deposition gun to print a foam statue with height more than 2 m (Barnett and Gosselin 2015). Izard et al. have tested large scale 3D printing with fully constrained CDPR as positioning device and have generated different print patterns with good accuracy and finish (Izard et al. 2017). Cable-driven robots can also be used in pick and place applications such as laying of bricks, beams, floor tiles, etc. and in assembly of window and door elements, with possible attachment of a serial robotic arm as end effector (Izard et al. 2017; Bosscher et al. 2007). Mattern et al. present a simulation study of automated bricking process using fully constrained cable-driven robot (Mattern et al. 2016). According to Bruckmann et al. (Bruckmann et al. 2018) the introduction of cable robots for construction, and the extension of Building Information Modelling (BIM) to define steps for robotic construction, will drastically improve automated construction in the near future. A detailed end effector design for brick laying can be found in this work.

The attractiveness of the underconstrained CDPR is that a large workspace free from the interference of cables is available. For example, in the case of the platform suspended using cables attached to the top of four poles erected at the corners of a construction site, the platform can move around without colliding, or cables interfering, with the objects in the site. This can be achieved by moving the platform to a higher plane first, and then traversing the length or breadth of the site. Even though the instability of the mechanism is a problem, it can be overcome to an extent if the nature of this instability is understood. This paper is an attempt towards this goal. The remaining part of this section briefs the existing literature on workspace of cable-driven robots. Sect. 2 describes the configuration and the mathematical model of the CDPR used for this study. Section 3 brings out the workspace analysis, where, the range of feasible orientations of the platform at different positions of the workspace is discussed. In Sect. 4, a simulation case study to highlight the usefulness of knowledge of feasible orientations in guiding the cable robot along a desired path is presented. In Sect. 5, the conclusions are drawn.

1.1 Present studies on workspace

The fully constrained CDPR has received more attention from researchers compared to underconstrained type. There are many studies on the workspace of fully constrained CDPR. The set of all poses of a CDPR, each of which can be achieved with positive forces in all cables is called a force-closure workspace (Pham et al. 2006). The set of poses that can be achieved with positive cable forces when a specific wrench set is acting on the mobile platform is termed a wrench-feasible workspace and the set of poses attainable for any wrench set is termed wrench-closure workspace (Gouttefarde and Gosselin 2006; Duan et al. 2015; Ouyang and Shang 2014). Gouttefarde and Gosselin present a study on the workspace of fully constrained planar four-cable CDPR (Gouttefarde and Gosselin 2004). This paper discusses several properties of constant orientation wrench-closure workspace. Gouttefarde and Gosselin extend this study for fully constrained planar CDPRs driven by more than four cables (Gouttefarde and Gosselin 2006). It details the wrench-closure workspace and presents an algorithm for finding the constant orientation wrench-closure workspace. Diao and Ma analyze the force-closure workspace of fully constrained CDPR driven by seven cables which are arranged arbitrarily in 3D space (Diao and Ma 2007). It describes the set of poses that satisfy the force-closure condition without any restriction on external wrench and acceleration of the manipulator. Stump and Kumar use convex analysis to derive expressions for workspace boundaries (Stump and Kumar 2006). Barrette and Gosselin analyze the workspace of cable-driven mechanism when it is confined to a plane (Barrette and Gosselin 2005). This work formulates a parametric representation for the two-dimensional workspace for any given orientation and acceleration of the platform/end effector. The work establishes that any position in this workspace can be attained with only three cables in tension.

Pham et al. use the ratio of cable tensions as a measure of quality of a solution to the inverse problem in fully constrained robots (Pham et al. 2009). The effect of connecting the base and the platform using springs can be seen in (Duan

et al. 2015; Gao et al. 2014), but generally this puts a limit to the size of the workspace. Mottola et al. discuss a special configuration of cables to form parallelograms so that the platform has translational motion (Mottola et al. 2019).

Researches on various aspects of direct and inverse problems of underconstrained CDPRs can be found in literature (Yanai 2001; Carricato 2013a, b; Abbasnejad and Carricato 2015). Xiang et al. describe a method of giving planned periodic motions to an underconstrained robot having a point mass and 3 degrees of freedom, so that, it can move beyond the static workspace (Xiang et al. 2020). Still, studies on the workspaces of underconstrained robots are comparatively less.

2 The CDPR model used

The space in which cable driven robots are intended to be used are typically cuboidal in shape, for example, a machine shop or construction site where CDPR is used to lift and transport objects. Here, a fixed frame X–Y-Z is assigned to the base with origin at one of the bottom corners of the space with the Z-axis pointing upwards. The moving platform or manipulator has the shape of a square lamina. Cables are attached to the four corners of the platform. A simplifying assumption is made that the center of gravity, G of the platform coincides with its geometric center. A frame x–y-z is attached to the platform with its origin at G. The x and y axes are parallel to the sides of the lamina. $\boldsymbol{B_i}$ is the vector from the center of gravity of the platform, G to the point of attachment of the ith cable to the platform, B_i, defined in the body-fixed frame x–y-z. The four upper corners of the workspace are designated as P_1, P_2, P_3 and P_4. The ith cable has one end attached to the platform at B_i and the other end to one of the upper corners of the workspace, P_i. The vector from B_i to P_i is $\boldsymbol{L_i}$. The cables are assumed to be massless, inextensible and perfectly flexible. The length of the ith cable, l_i, is the magnitude of $\boldsymbol{L_i}$. The position vector of P_i in the fixed frame X–Y-Z is $\boldsymbol{P_i}$ and the position vector of G in the fixed frame is $\boldsymbol{R}$. The orientation of the platform is defined using Euler angles. Initially, the axes of the body co-ordinate system are parallel to the corresponding axes of the fixed co-ordinate system. Any orientation can be obtained by rotating an angle θ about the X-axis, followed by a rotation of φ about y-axis, followed by a rotation of ψ about z-axis. The cable-driven parallel robot is shown in Fig. 1. The static equilibrium is dictated by the cable forces and the weight of the platform. As the cables can only pull and not exert push, the forces in the cables are always non-negative. When the force in a cable is zero, that cable is inactive at that instant and the mobile platform is controlled by the other cables.

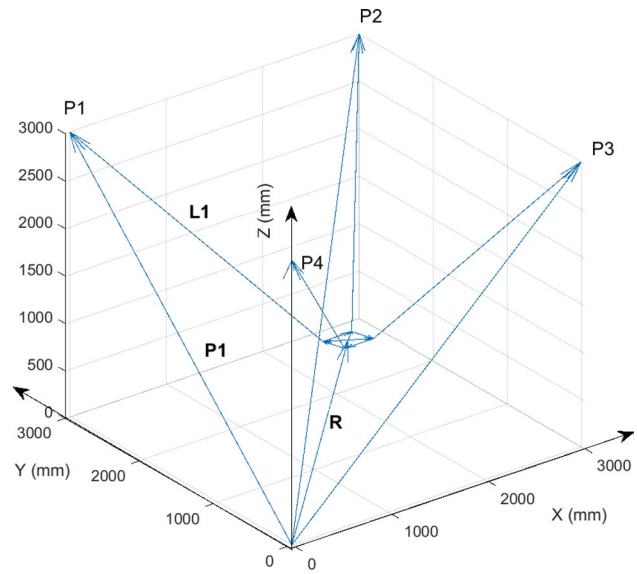

Fig. 1 Configuration of the cable-driven parallel robot. $P1$, $P2$, $P3$ and $P4$ represent the cable attachment points. $\boldsymbol{P1}$, $\boldsymbol{L1}$ and $\boldsymbol{R}$ (in bold) represent typical position vectors

For any given position of the centre of the platform, $\boldsymbol{R} = \begin{bmatrix} X & Y & Z \end{bmatrix}^{\mathrm{T}}$ and orientation of body co-ordinate frame $\boldsymbol{\Theta} = \begin{bmatrix} \theta & \varphi & \psi \end{bmatrix}^{\mathrm{T}}$, $\boldsymbol{L_i}$ are obtained by solving the loop-closure equations,

$$\mathbf{R} + \mathbf{AB_i} + \mathbf{L_i} - \mathbf{P_i} = 0, i = 1, \ldots 4.$$

where, $\mathbf{A}$ is the rotation matrix from body fixed frame x–y-z to fixed frame X–Y-Z. The set of string lengths [l_1, l_2, l_3, l_4] are obtained as,

$$l_i = |\mathbf{L_i}|, i = 1, \ldots 4.$$

For validating the computational model, a physical model was built. The dimensions of the physical model were selected such that the parameters are measurable and at the same time it is compact. This physical model has the dimensions as: the position vectors $\boldsymbol{P_1}$, $\boldsymbol{P_2}$, $\boldsymbol{P_3}$ and $\boldsymbol{P_4}$ are $\begin{bmatrix} 0 & 3 & 3 \end{bmatrix}^{\mathrm{T}}$, $\begin{bmatrix} 3 & 3 & 3 \end{bmatrix}^{\mathrm{T}}$, $\begin{bmatrix} 3 & 0 & 3 \end{bmatrix}^{\mathrm{T}}$ and $\begin{bmatrix} 0 & 0 & 3 \end{bmatrix}^{\mathrm{T}}$ metres respectively; the position vectors $\boldsymbol{B_1}$, $\boldsymbol{B_2}$, $\boldsymbol{B_3}$ and $\boldsymbol{B_4}$ defined in body fixed co-ordinate system x–y-z are $\begin{bmatrix} -0.150 & 0.150 & 0 \end{bmatrix}^{\mathrm{T}}$, $\begin{bmatrix} 0.150 & 0.150 & 0 \end{bmatrix}^{\mathrm{T}}$, $\begin{bmatrix} 0.150 & -0.150 & 0 \end{bmatrix}^{\mathrm{T}}$ and $\begin{bmatrix} -0.150 & -0.150 & 0 \end{bmatrix}^{\mathrm{T}}$ metres respectively; and the mass of the platform is 300 g. The computation model presented here also has the same dimensions.

A position and orientation of the platform, specified by $\mathbf{R}$ and $\boldsymbol{\Theta}$, is possible if the platform is in equilibrium with the forces in all the strings non-negative. When the

platform is in equilibrium under the action of the forces in the strings and gravity,

$$\mathbf{KF} = \mathbf{W}$$

where, $\mathbf{K} = \begin{bmatrix} \mathbf{U}_1 & \mathbf{U}_2 & \mathbf{U}_3 & \mathbf{U}_4 \\ \mathbf{M}_1 & \mathbf{M}_2 & \mathbf{M}_3 & \mathbf{M}_4 \end{bmatrix}$, $\boldsymbol{U}_\mathbf{i}$ is the unit vector along direction of $\boldsymbol{L}_\mathbf{i}$, and

$$\mathbf{M_i} = (\mathbf{AB_i}) \times \mathbf{U_i};$$

$\mathbf{F} = \begin{bmatrix} F_1 & F_2 & F_3 & F_4 \end{bmatrix}^T$, where, F_i is the force in the *ith* string in Newton, and.

$\mathbf{W} = \begin{bmatrix} 0 & 0 & mg & 0 & 0 & 0 \end{bmatrix}^T$, mg being the weight of the platform in Newton.

3 Analysis of the workspace

At every location in the workspace to which the centre of the platform is moved to, there is a range of inclinations possible. Knowing the possible range of inclinations corresponding to every position will be useful in guiding the platform.

Here, it is not attempted to find the positions and orientations where the platform is perfectly in equilibrium. Instead, the possible poses of the platform, where, the net force in X direction, Y direction and Z direction, and the net moment of force about vertical axis through G, are each equal to zero; and the components of moment of forces about G, parallel to X direction and Y direction are within specified limits, are found. This condition can be expressed as follows:

$$\mathbf{K}'\mathbf{F} = \mathbf{W}'$$

where, $\mathbf{K}'$ and $\mathbf{W}'$ are obtained by deleting the 4th and 5th rows of $\mathbf{K}$ and $\mathbf{W}$ respectively, representing the balance of forces in the X, Y and Z directions, and the balance of moments about the vertical axis through G. On solving this equation, the set of forces in the strings, $\mathbf{F}$ is obtained. With $\mathbf{M}'$, a subset of $\mathbf{M}$, containing its 1st and 2nd rows, the inequality $\left|\mathbf{M}'\mathbf{F}\right| < m$, represents the condition that, the components of moment of forces about G, parallel to X and Y directions, are each less than a specified value m. The set of poses of the platform that satisfy these conditions are mapped.

Even though, the poses found are not in complete equilibrium, for a moving platform they indicate the trajectory where it is nearly stable. At any position, an orientation away from this possible range can be achieved only with an appropriate wrench set acting on the platform. While guiding the platform, if an orientation away from the feasible range is attempted, the lengths calculated for this pose will result in a considerable drift in the position of the centre of the platform. Therefore, knowledge of this feasible range of orientations will help in limiting the drift in position.

The range of angles, $\mathbf{\Theta} = \begin{bmatrix} \theta & \varphi & \psi \end{bmatrix}^T$ possible, at typical positions in the workspace, where string forces remain non-negative and the net force in any direction and moment of forces about vertical axis through G are all equal to zero, and the components of moment of forces about G, parallel to X and Y directions are within specified limit m, is presented as follows:

Choosing a typical point for the center of the platform, such as, $X = 0.5$ m, $Y = 1$ m, $Z = 2$ m, which is not at any plane of symmetry, the range of possible orientations are plotted in Fig. 2, with horizontal and vertical axes representing ψ and φ respectively. The value of m is taken as 0.1 Nm, i.e., for these poses the components of moment about G, parallel to X and Y directions are each less than 0.1 Nm in magnitude. The angle θ is kept constant at $-20°$. The value of φ is incremented by 0.2° in the range $-90°$ to $+90°$ and ψ incremented by 0.1° in the range $-90°$ to $+90°$ for finding the feasible points in ψ-φ space. Salient values are given in Table 1 which shows the range of feasible values for ψ, for different values of φ, with θ remaining same at $-20°$. It also shows the cable forces at these extreme values of ψ. The value that limits the range, whether cable force becoming negative or moment about an axis exceeding the specified value, is also shown in the table. Some of the values are explained as follows: when θ is set as $-20°$ and φ takes the value of 35°, ψ can vary from 4.25° to 5.83° with all cables in tension. Beyond 4.25° the moment of forces about X direction through G, M_x becomes greater than 0.1 Nm; and beyond 5.83° cable force F_2 becomes negative. When φ

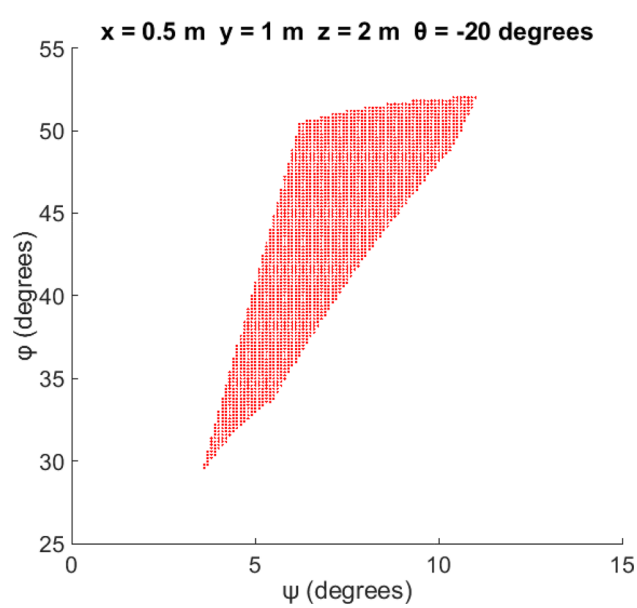

Fig. 2 Possible orientations at a position when θ remains fixed; m = 0.1 Nm

Table 1 The range of values of angle ψ for different values of φ at a position in the workspace when θ remains fixed

Position: $X = 0.5$ m, $Y = 1$ m, $Z = 2$ m; $\theta = -20°$; m = 0.1 Nm

φ	Lower limit of ψ	F_1	F_2	F_3	F_4	Limited by value of	Upper limit of ψ	F_1	F_2	F_3	F_4	Limited by value of
30	3.61	1.7526	0.5971	0.6672	2.4708	$M_x = 0.0994$	3.78	1.8221	0.4899	0.7577	2.4278	$M_y = 0.0999$
35	4.25	1.7692	0.5944	0.6995	2.4703	$M_x = 0.0994$	5.83	2.1484	0.0014	1.2018	2.2340	$F_2 = 0.0014$
40	4.89	1.7739	0.6082	0.7195	2.4757	$M_x = 0.0998$	7.26	2.1560	0.0007	1.2362	2.2350	$F_2 = 0.0007$
45	5.53	1.7730	0.6285	0.7355	2.4829	$M_x = 0.0995$	8.90	2.1609	0.0001	1.2723	2.2353	$F_2 = 0.0001$
50	6.14	1.7663	0.6560	0.7462	2.4923	$M_x = 0.0997$	10.60	2.1478	0.0245	1.2881	2.2447	$M_x = -0.0998$

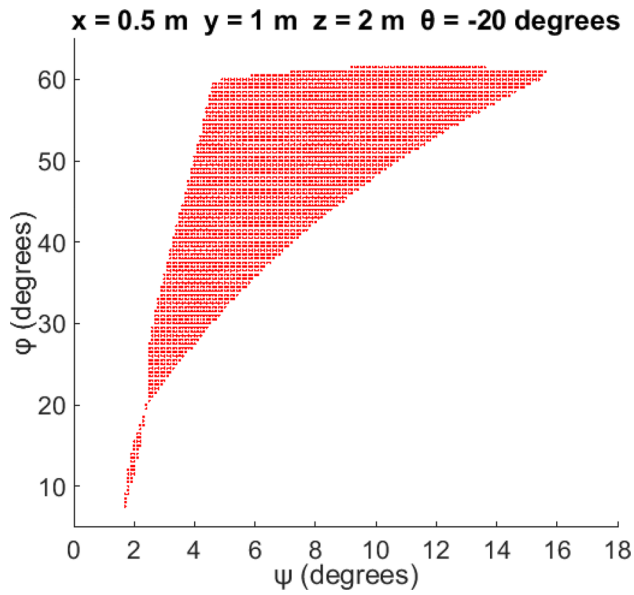

Fig. 3 Increase in range of angles when limiting value of moment is increased; $m = 0.2$ Nm

is set as 40°, the feasible range of ψ becomes 4.89° to 7.26°. Beyond 4.89°, again the moment of forces about X direction, M_x exceeds the limit and beyond 7.26° F_2 becomes negative.

When m, the maximum value for components of moment about G along X and Y directions, is increased, it can be observed that the range of possible orientations increases. Figure 3 shows the plot with maximum limit for moment, $m = 0.2$ Nm, the horizontal and vertical axes representing the angles ψ and ψ, and the position of the centre of the platform and angle θ remaining the same as before, i.e., [X, Y, Z] = [0.5, 1, 2] m and θ = − 20°.

The range of feasible orientations for different values of θ, with the centre of the platform at the same position [X, Y, Z] = [0.5, 1, 2] m, is shown in Fig. 4 as a series of plots with each plot having a fixed value of θ and with ψ and φ represented along the horizontal and vertical axes. Limiting value for moment is taken as $m = 0.1$ Nm. It can be seen that for this position the range of inclinations is wider when θ is slightly negative, i.e., when the upper surface of the platform is tilted towards the centre of the workspace. At $\theta = 30°$, there is no feasible solution.

The variation in the range of possible orientations with variation in height is shown in Fig. 5, where the X and Y co-ordinates of the centre of the platform is kept constant at 0.5 m and 1 m respectively and the angle θ is kept fixed at − 20°. The height of the centre of the platform, Z is held at different values 2 m, 1.5 m, 1 m and 0.5 m. When the platform is lowered, as in the case when $Z = 0.5$ m, it can be seen that the platform can be tilted in a wider range. As the height is increased, the cables become tauter and the range decreases.

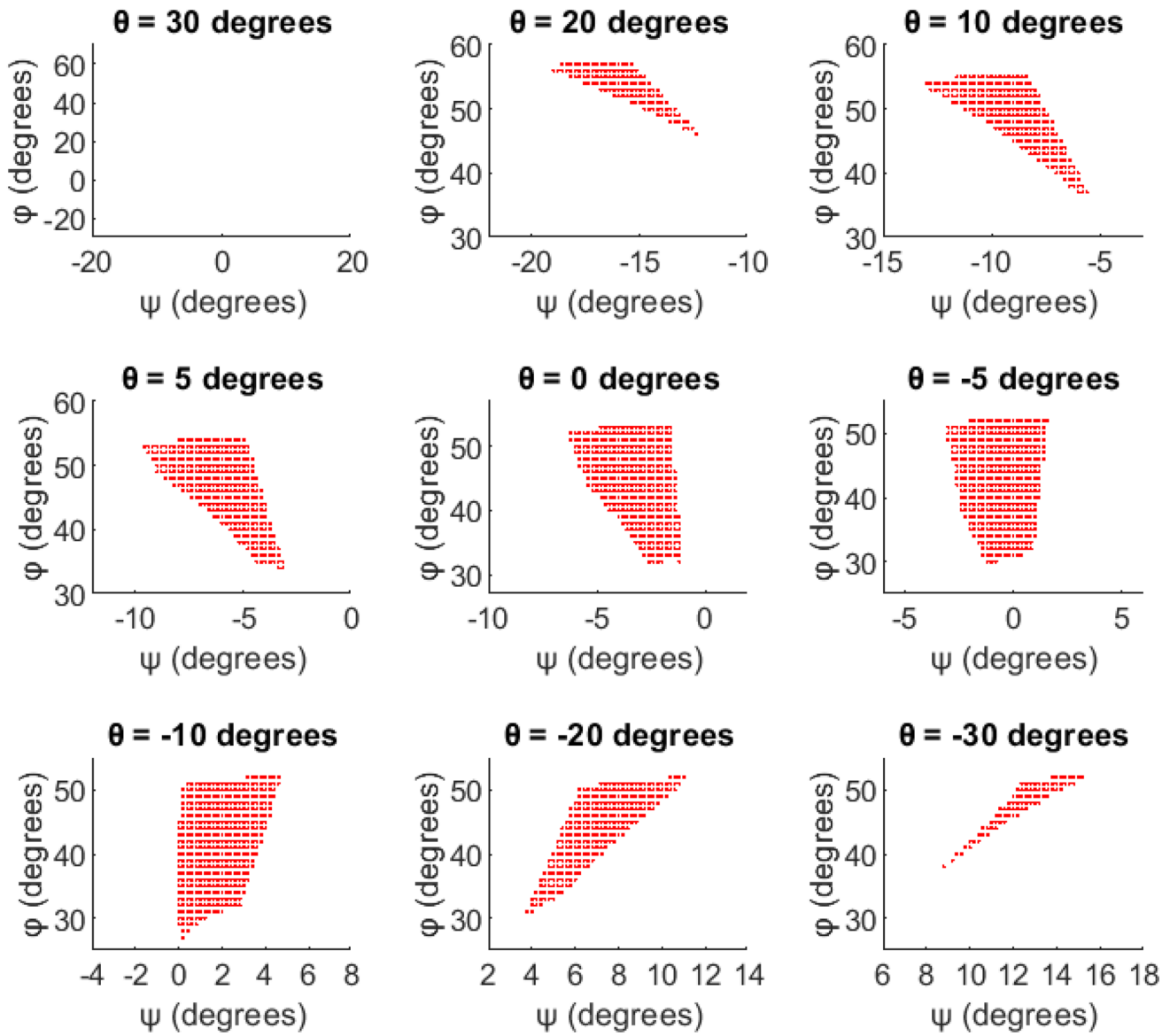

Fig. 4 Range of orientations for different values of θ; position $[X, Y, Z] = [0.5, 1, 2]$ m and $m = 0.1$ Nm

Now, the possible orientations when the centre of the platform is moved to different positions keeping the height constant at 2 m, and keeping the angle θ constant at $-20°$, are shown in Fig. 6. It can be observed that as the value of X increases from 0.5 m to 1.5 m, i.e., as the platform move towards the center of the workspace, the value of φ decreases. The angle φ is the rotation about positive y direction of the body-fixed frame x-y-z of the platform. This decrease in angle φ means that the platform becomes more and more horizontal. For the given value of angle θ, there are no solutions at two of the locations as can be seen from the figure.

It can be seen that with a wrench set that consists only the cable forces and the weight of the platform, the feasible range of orientations of the mobile platform at any position is limited to a small range. Inclinations of the platform away from this range are possible only with appropriate wrench sets. If cable lengths calculated for an infeasible orientation are applied, the platform will settle into an equilibrium position with a probable drift in the position from that specified. Thus knowledge of the feasible range of orientations will be useful to guide the mobile platform along a desired path.

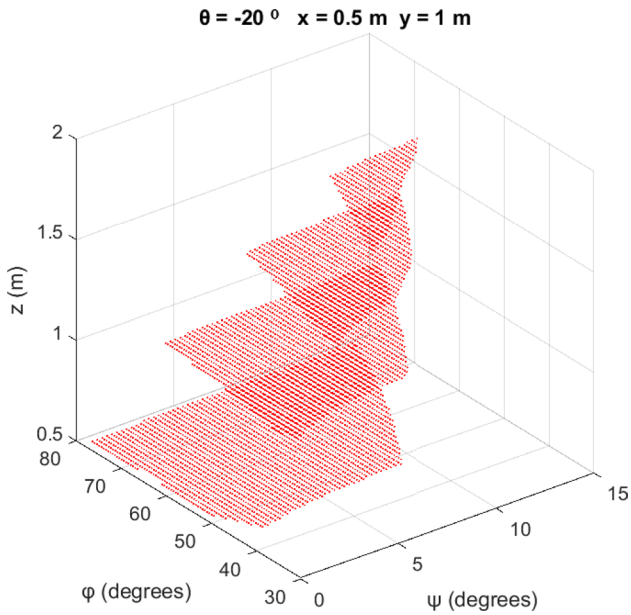

Fig. 5 Change in range of possible orientations with change in height of the centre of the platform; [*X, Y*] = [0.5, 1] m, $\theta = -20°$, $m = 0.1$ Nm. Height *Z* at 0.5 m, 1 m, 1.5 m and 2 m

4 A case study using simulation

A simulation study of the cable-driven robot as it transports objects in a hypothetical construction site is conducted, with focus on the adherence to the desired path. An end effector in the form of a gripper is attached to the moving platform to carry objects. The study tests the usefulness of knowing feasible orientations, while guiding the robot along any path. The site has a rectangular shape with length 22 m and width 15 m. The moving platform is suspended using four cables, attached to poles erected at the corners of the plot, with the attachment points at a height of 10 m from the ground. The platform is a square lamina of sides 900 mm each. The platform, end effector and the object being lifted together has a mass of 570 kg. Figure 7 shows a 3D model of the construction site and the cable-driven robot transporting an object.

The moving platform carrying a payload is intended to be guided with the geometric center of the platform moving along a straight line. Three points *A*, *B* and *C* which lie on the straight line path, are taken as the test points. The points have co-ordinates (11, 10, 7) m, (14, 10, 7) m and (17, 10, 7) m respectively with respect to a frame which has origin at a corner of the plot, *X*-axis along its length, *Y*-axis along width and *Z*-axis pointing upwards. Figure 8 shows the intended straight path of the platform and the positions of points *A*, *B* and *C*. At each of these points, one feasible orientation, represented by the three Euler angles, is selected, and the required cable lengths are determined by the method described in Sect. 2. The simulation is done using NX9 software package. The cable lengths corresponding to a pose, are given as input and the model is solved for static equilibrium, with gravity enabled. From the solution it is observed whether or not the centre of the platform remains at the corresponding test point. Only a static simulation is done and the effect of inertia is not studied.

The end effector is a gripper for holding the objects to be lifted. If the gripper is rigidly attached to the moving platform, due to inclination of the platform, the weight of the payload and the gripper will produce a torque about the centre of the platform. One method to overcome this is by using a ball and socket joint for connecting the gripper to the platform. Here, the centre of spheres of the ball and socket joint is designed to coincide with the geometric centre of the platform, which also is the geometric centre of the cable attachment points on the platform. Thus, if the payload is gripped by its centre of gravity, the weight of the gripper and payload will always pass through the centre of the platform, and the equations used in the mathematical model in Sect. 2 will remain valid. The disadvantage of this design is that, there will not be any control over the orientation of the object about the vertical axis.

The feasible orientation selected for point *A* is (10, 0, 0) ° and the set of cable lengths corresponding to this pose is (11.858, 11.858, 14.564, 14.564) m. A feasible orientation (10, − 4, 0) ° is chosen for point *B*, with (14.597, 9.284, 12.551, 16.874) m being the set of cable lengths for achieving this pose. For point C, the feasible orientation selected is (10, − 12, 2)° and the set of cable lengths is (17.424, 7.051, 10.993, 19.394) m. The maximum limit for *X*-component and *Y*-component of the unbalanced moment of forces about G, the centre of the platform is taken as 200 Nm, for considering an orientation as feasible. The effect of giving a set of cable lengths, corresponding to an infeasible orientation, as input, is also tested at point C. The orientation (0, 0, 0) ° which is infeasible at point C is selected and the set of cable lengths corresponding to this pose is (17.424, 7.100, 10.996, 19.342) m.

The results of the simulation are given in Fig. 9. For the first three cases where cable lengths corresponding to feasible orientations are given as input, the centre of the platform stays close to the corresponding test point. These results are shown in Fig. 9 (a)–(c). For the first simulation, the desired point is *A* (11, 10, 7) m, and the simulation result for position of platform centre is (11.0000, 10.0017, 6.9999) m. The shift in the position of platform centre from the intended position is (0, 1.7, − 0.1) mm. For the second simulation the desired position is *B* (14, 10, 7) m and the simulation result is (13.9975, 10.0038, 6.9999) m. The shift in position is (− 2.5, 3.8, − 0.1) mm. And for the third input, the intended position is *C* (17, 10, 7) m and the position obtained in

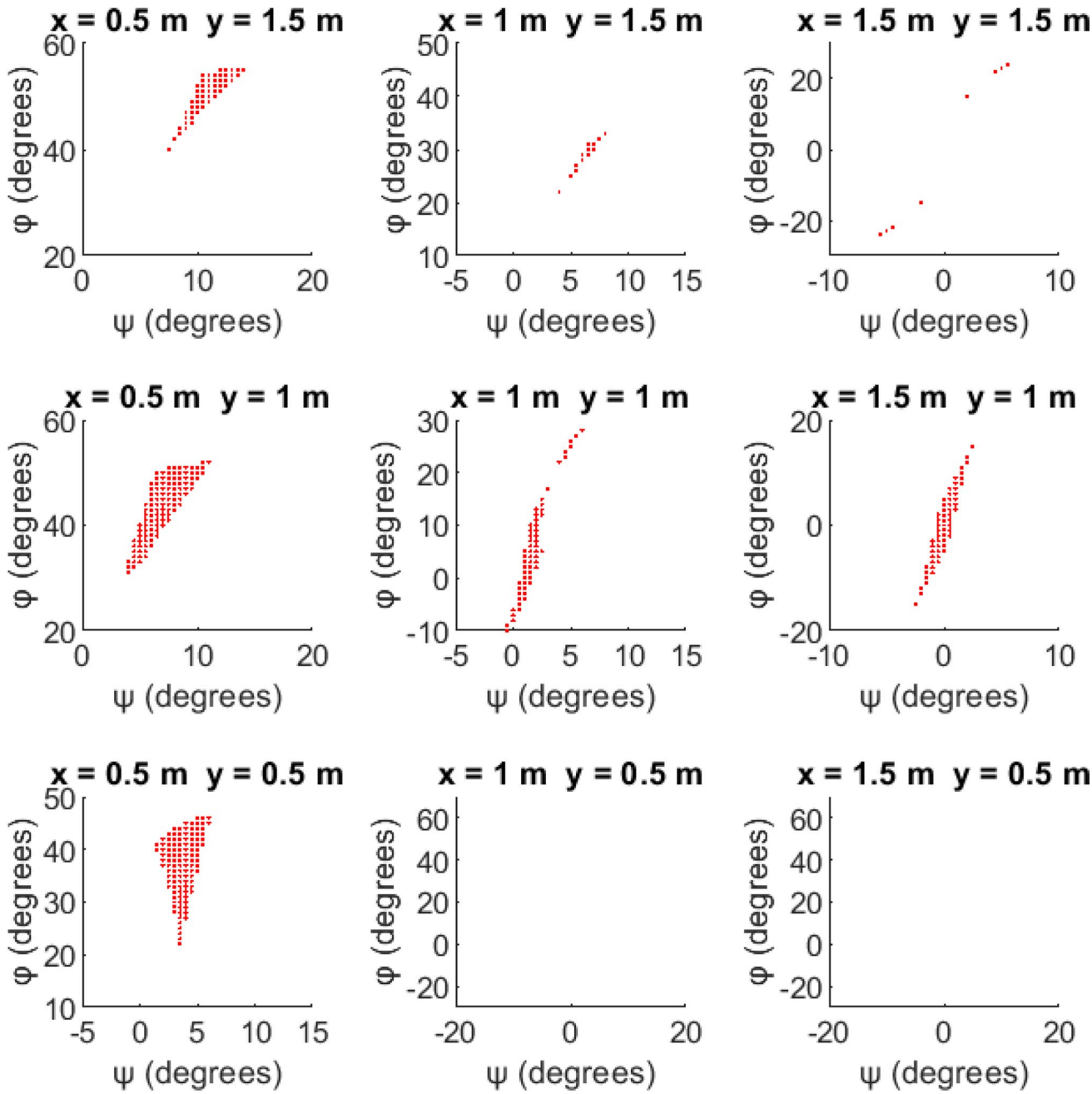

Fig. 6 Range of possible orientations, at different positions of the workspace with height and angle θ kept constant; $Z = 2$ m, $\theta = -20°$, $m = 0.1$ Nm

simulation is (16.9931, 10.0040, 7.0005) m. The shift in position is (− 6.9, 4.0, 0.5) mm. It should be remembered that, the sets of cables lengths given as inputs to the simulation, were determined from poses which were not fully in equilibrium. Unbalanced couples may exist at these feasible orientations. For the pose defined at point A, the resultant moment of forces, given by the product **MF** as explained in Sect. 3, is $\left[-86.3\ \ 0\ \ 0\right]^{\mathrm{T}}$ Nm. At B, it is $\left[-114.5\ \ -64.7\ \ 0\right]^{\mathrm{T}}$ Nm and at C, $\left[17.3\ \ -107.1\ \ 0\right]^{\mathrm{T}}$ Nm. The presence of these couples can be the cause for this small deviation in equilibrium position.

The result for the fourth case, where, cable lengths corresponding to an infeasible orientation are given as input, is given in Fig. 9 (d). It is observed that the platform centre deviates from the intended position. The desired position is C (17,

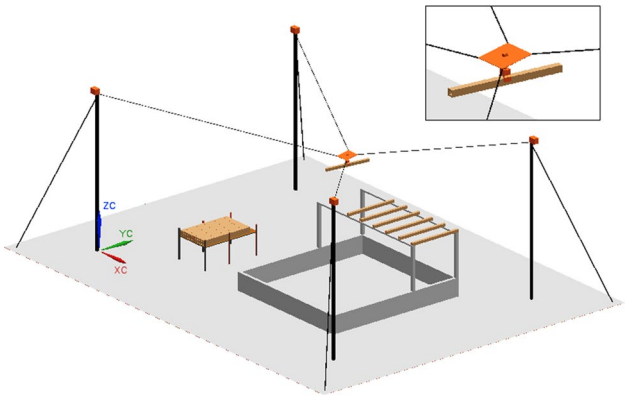

Fig. 7 A representation of cable-driven robot employed in a construction site for lifting and transferring objects. The close-up view of the gripper holding the object is shown in the inset

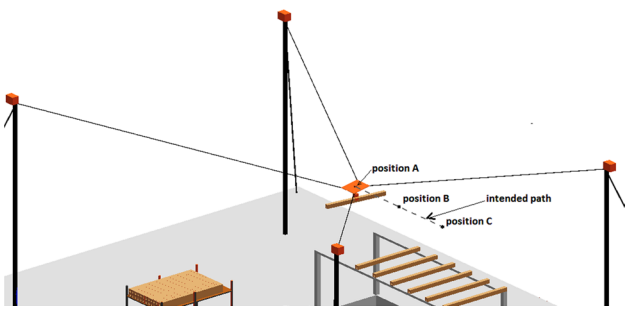

Fig. 8 An image of the workspace of cable-driven robot and the straight line path along which the centre of the platform is intended to be guided. The three test positions ***A***, ***B*** and ***C*** are also shown

10, 7) m and the obtained position is (16.9580, 9.9969, 6.9634) m. The drift of the centre of the platform from the desired position is (− 42, −3.1, − 36.6) mm. This can be attributed to the existence of comparatively large unbalanced moment of $\begin{bmatrix} 892.3 & -1431.1 & 0 \end{bmatrix}^T$ Nm corresponding to this pose.

5 Conclusions

Cable-driven robots offer the advantages of large workspace, high payload to weight ratio and easiness in re-configuring. These advantages favour their use in construction tasks like positioning device for concrete extruder nozzle in 3D printing of buildings and structures, lifting and transferring materials, and laying of beams, bricks, etc. In literature, both versions of CDPRs, underconstrained ones and fully constrained ones, can be seen recommended for use in large scale additive manufacturing (Bosscher et al. 2007; Barnett and Gosselin 2015; Izard et al. 2017). These two types of CDPRs differ in their abilities. While fully constrained CDPRs offer high stiffness for end effector pose, against reactive forces and torques resulting from various tasks, their lower cables can interfere with objects and thus limits the usable workspace. Some of the researchers who employed fully constrained CDPR in additive manufacturing of buildings, have overcome this difficulty by progressively raising the attachment points of lower cables as the structure is built layer by layer (Bosscher et al. 2007; Izard et al. 2017). The advantage of underconstrained CDPR is that they have cables attached only to the upper side of workspace and therefore they can move without cables interfering with objects in a site or reach inside a constructed building.

The disadvantages of underconstrained CDPRs are their instability against forces on the end effector and the limited range of orientations possible. This paper has attempted to map the feasible range of orientations at different positions in the workspace, so that, it can be made use of in tasks that do not encounter considerable reactive forces or torques other than gravity force.

The range of feasible orientations of an underconstrained CDPR, with only the cable forces and gravity force acting on it, are analyzed. It is not attempted to find the range where the platform is in complete static equilibrium; rather, the range of poses with, net force in any direction equal to zero, moment about vertical axis through the centre of the platform equal to zero, and the moment about the other two axes through centre limited to a small value, are found. The range of orientations at different locations along X and Y directions, and at different heights are mapped. A simulation study of equilibrium of the cable-driven robot at different positions along a prescribed path, as it transfers a payload in a hypothetical construction site is done. It is found that the moving platform follows the path when feasible orientations are used and when infeasible orientation is attempted the centre of the platform moves away from the intended path. Thus the importance of knowing the feasible range of orientations is highlighted.

Furthermore, it can be stated that, to tap the advantages of both underconstrained and fully constrained types of CDPRs, with the same hardware, and with suitable changes in the control algorithm, the CDPR can be changed between fully constrained and underconstrained types. With fully constrained version for tasks requiring stiffness against end effector forces and underconstrained version for moving the end effector amidst obstacles, CDPRs offer a promising choice

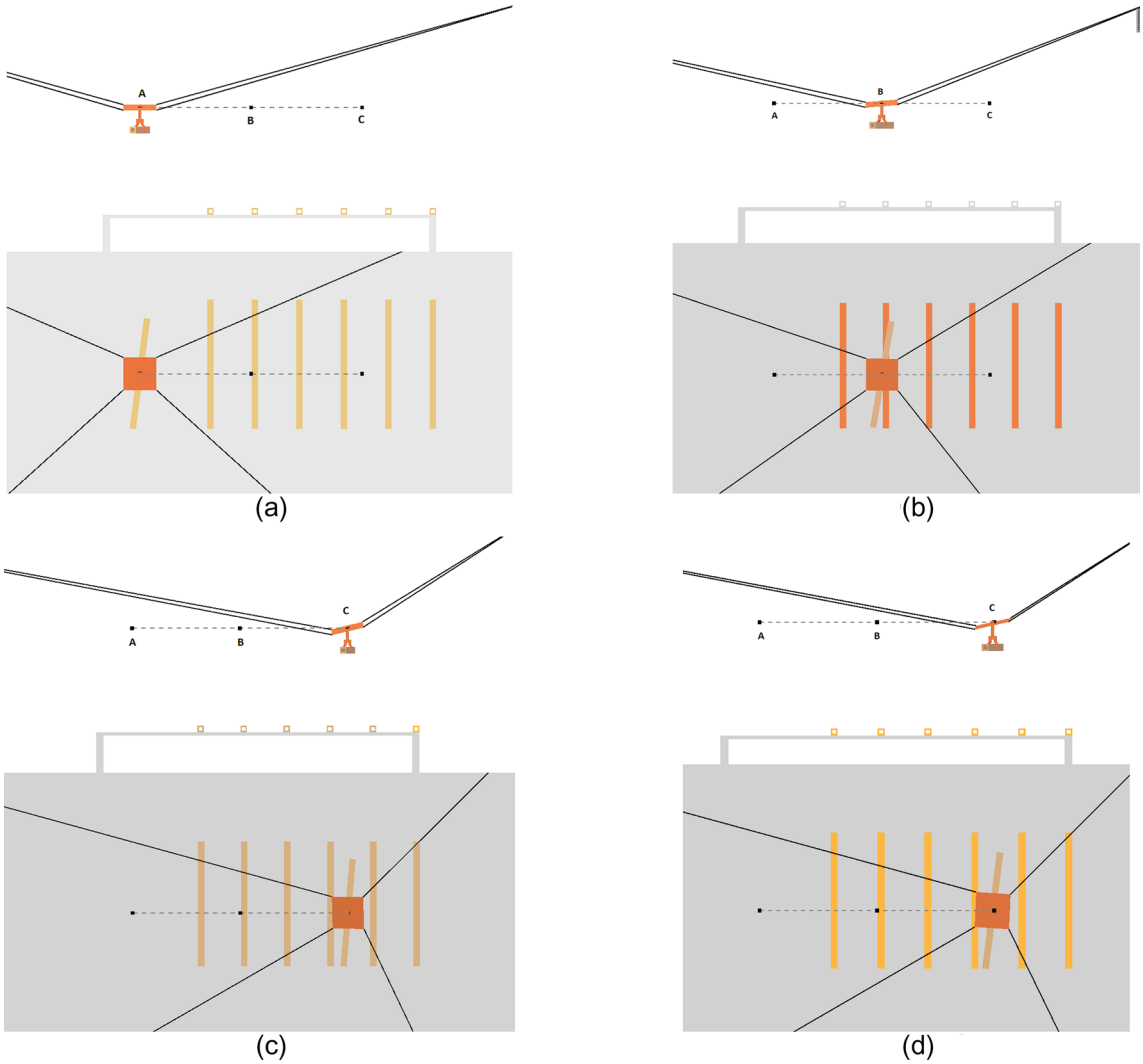

Fig. 9 Equilibrium positions and orientations of the mobile platform carrying payload obtained by simulation for different sets of lengths as input. Each result is shown in front view and top view. The path and test points can also be seen. **a** When a set of lengths corresponding to a feasible orientation at point ***A*** is given as input. **b** Resulting pose when cable lengths corresponding to a feasible orientation at point *B* are given. **c** When cable lengths corresponding to a feasible orientation at point *C* are given as input. **d** Result for a set of lengths corresponding to an infeasible orientation at point *C*.

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Declarations

Conflict of interest On behalf of all authors, the corresponding author states that there is no conflict of interest.

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